Finite label costs for certifying best-of-N risk

Larry Mai (project owner), with substantial Codex assistance

Research manuscript, 27 October 2026. This document consolidates the finite-cost results in the public research artifact at commit `765ac51` [1]. It reports no new experiment.

Contribution disclosure. Larry Mai commissioned this project. Codex developed the theory, software, literature assessment, numerical checks, and manuscript; separate agent contexts reviewed the work. Larry has not personally checked the results. There has been no independent human verification or peer review. The mathematical statements are contribution candidates; scientific priority and practical model benefit remain unestablished.

Abstract

Consider a fixed collection of ranked candidate pools and a declared family of best-of-N selectors. An auditor purchases correct, fixed losses and may certify that every declared selector has risk at most a threshold. We study the least worst-case expected number of distinct labels on tables separated from that threshold by a supplied margin. Validity is required on every fixed table, and completion on every safe table, including the boundary. For every finite pool size, prompt count, and nonempty budget grid, the cost is characterized up to universal constants by the smaller of a deterministic prefix cost and a clipped envelope sum. A construction using classical likelihood-ratio tests and a shared label cache attains a factor-17 upper bound on geometrically separated grids. The lower proof combines a common safe prior across diffuse rank scales with exact-count priors for finite saturation. For a full dyadic grid and a particular shrinking-margin sequence, the matching cost is of order $m\log\log(m)/\log(m)$, below census order. The constants are loose; a separate finite-instance certificate illustrates that practical savings can remain modest. These are fixed-table query-cost results, not measured LLM improvements or guarantees about graders, new prompts, or deployment.

1. The question and the contribution candidate

A higher search budget changes which ranked candidates are likely to be selected. An audit of several budgets can reuse a purchased label, but optimizing a sampling weight or an estimator's variance need not minimize the labels needed to certify a threshold decision. We ask how expensive that decision must be when the auditor can choose queries adaptively and share observations across every declared budget.

The decision is deliberately specific: issue one certificate asserting that the entire declared family has risk at most τ, or abstain. It does not return an arbitrary selected subset of safe budgets. This distinction matters both for error accounting and for comparisons with simultaneous confidence bands.

Finite-pool winner-rank evaluation is established [2]. Envelope acquisition and reusable best-of-N inference have close predecessors [3,4]. Clipped inclusion probabilities, deterministic inspection of heavy items, and truncation once little weight remains are also established [5,6]. The candidate contribution is the complete finite characterization below for this particular rank family and decision, together with an upper construction that charges every acquired label. It does not introduce a new generic sampling or sequential-testing principle.

The proof is provided in three appendices. Appendix A proves the all-grid lower bound; Appendix B proves the cached and grouped upper bounds; Appendix C supplies the finite-population and residual-concentration lemmas needed by both. The main text explains the quantifiers, geometry, and consequences before those technical arguments.

2. Fixed experiment and exact cost objective

There are $P \geq 1$ equally weighted prompt pools, each with $m \geq 1$ distinct items in a fixed total rank order, with rank 1 best. The order includes a declared rule for ties. A query of item (p, r) reveals its fixed, correct loss $y_{p,r} \in [0, 1]$ at unit cost. No labels are supplied for free, and no trusted cross-item relationship reveals another item's loss. Computation and auxiliary randomness are uncharged.

Fix a nonempty grid $\mathcal{K} \subseteq \{1, \ldots, m\}$. At budget n, selection within each pool returns the best-ranked item in a uniform n-subset drawn without replacement. The winner law and risk are

$$w_n(r) = \frac{\binom{m-r}{n-1}}{\binom{m}{n}}, \qquad a_n(p, r) = \frac{w_n(r)}{P},$$

$$R_n(y) = \sum_{p=1}^{P} \sum_{r=1}^{m} a_n(p, r) y_{p,r}.$$

Impossible binomial choices are zero. Each target row sums to one. These are conditional risks on the recorded pools; the fixed losses need not be independent across prompts or candidates. Selecting the best of fresh independent generations would be a different experiment.

An admissible auditor may randomize, select unseen items adaptively, reuse cached labels, and abstain. Its certificate asserts that $R_n(y) \leq \tau$ for every originally declared n. On every fixed table with some $R_n(y) > \tau$, the probability of ever issuing that certificate is at most δ. On every fixed safe table it must certify after finitely many queries almost surely, including when a risk equals τ. These requirements apply to all of $[0, 1]^{Pm}$. Exact evaluation after a full census shows that the admissible class is nonempty.

For a supplied design margin ε, define

$$\mathcal{Y}_\varepsilon = \{y : \max_{n \in \mathcal{K}} R_n(y) \leq \tau - \varepsilon\},$$

$$C^*(\varepsilon) = \inf_A \sup_{y \in \mathcal{Y}_\varepsilon} \mathbb{E}_y T_A(y).$$

Here T_A counts distinct labels bought before certification. One auditor serves the entire margin class. The margin can guide its design; it is not trusted evidence that an unknown table is safe. Expectations concern audit randomization on a fixed table. The comparator permits correlated sampling and even empty-query certification lotteries, provided global validity holds. It places no computational restriction on the auditor.

3. The finite-cost formula

Write $a = \min \mathcal{K}$, $L = \log(1/\delta)$, and

$$H = \frac{\tau L}{\varepsilon^2}, \qquad e_r = \max_{n \in \mathcal{K}} w_n(r).$$

Let $q_a(k)$ be the smallest budget's probability of selecting beyond rank k, and let k_ε be the first prefix leaving at most ε such mass:

$$q_a(k) = \frac{\binom{m-k}{a}}{\binom{m}{a}}, \qquad k_\varepsilon = \min\{0 \leq k \leq m : q_a(k) \leq \varepsilon\}.$$

Define the clipped sum and the final functional by

$$G = \sum_{r=1}^{m} \min\{P, He_r\}, \qquad F = \min\{Pk_\varepsilon, G\}.$$

Theorem 1 (all finite grids). If

$$0<\tau\le\tfrac{1}{8},\qquad 0<\varepsilon\le\tfrac{\tau}{16},\qquad 0<\delta\le\tfrac{1}{4},$$

then for every finite P, m and every nonempty declared grid,

$$\tfrac{F}{2^{38}}\le C^{*}(\varepsilon)\le 2^{16}F.$$

If consecutive declared budgets satisfy $n_{j+1}\ge 2n_j$, then $C^{*}(\varepsilon)\le 17F$. Singletons satisfy this spacing condition. There is no density, replication, or minimum-budget restriction in Theorem 1. Its universal constants are deliberately loose and do not claim an optimal leading coefficient.

The two terms in F describe different finite effects. Since the rows and envelope decrease with rank,

$$G=\min_{0\le k\le m}\{Pk+H\sum_{r>k}e_r\}.$$

Increasing k changes the expression by $P-He_{k+1}$; its nondecreasing increments prove the identity. The clipped term pays once for heavy items, while assigning smaller costs to lighter items. Algebraically, it is the expected sample size of classical inclusion probabilities $\min\{1, He_r/P\}$ [5]. This correspondence alone does not prove a certification-cost optimum.

The prefix term has a deterministic interpretation. Query every prompt at ranks $1,\dots,k_\varepsilon$. All larger budgets have no more tail mass than budget a, so the known prefix plus $q_a(k_\varepsilon)$ bounds every risk. On $\mathcal{Y}_\varepsilon$ all those bounds pass. If they fail on another table, acquire the remaining relevant labels and evaluate the family exactly. This is globally valid and completes on every safe table. Its margin-class cost is at most Pk_ε, even as δ tends to zero. Tail truncation is an established weighted-query idea [6]; the theorem's additional claim is its role in the matching finite formula.

4. A cached likelihood construction

The separated-grid upper bound has a simple description using classical conservative likelihood-ratio tests [7]. Put $\theta=\tau-\varepsilon$ and

$$u=\log\tfrac{1-\theta}{1-\tau},\qquad d=\mathrm{kl}(\mathrm{Ber}(\theta),\mathrm{Ber}(\tau)),$$

$$c=\tfrac{L+u}{d},\qquad B_c=\sum_{r=1}^{m}\min\{P, c\sum_{n\in\mathcal{K}}w_n(r)\}.$$

Theorem 2 (cached upper bound). For any grid, whenever $0<\varepsilon<\tau<1$ and $0<\delta<1$, an admissible auditor has margin-class expected distinct-label cost at most B_c. Consequently, $C^{*}(\varepsilon)\le\min\{Pk_\varepsilon, B_c\}$.

Process the declared budgets in a fixed order, with an independent iid proposal stream of item law a_n for each budget. All streams share one label cache. Buy the label on the first proposal of an item. On *every* proposal, including a cache hit, generate a fresh independent coin $B\sim\mathrm{Ber}(y_{p,r})$. For a fixed table the coins within the current stream are iid with mean $R_n(y)$. The log-likelihood update for a coin value B is

$$B\log\tfrac{\theta}{\tau}+(1-B)\log\tfrac{1-\theta}{1-\tau}.$$

Pass a budget when the cumulative sum reaches $I_\cdot$. Also pass by exact evaluation if all its positive-support labels have been acquired and its risk is at most τ; terminate without certification if exact evaluation finds it unsafe. Certify the family only after every budget passes. The finite iid proposal stream visits its entire positive support almost surely, so this exact check supplies completion on boundary-safe tables. Every first acquisition remains charged.

On a fixed unsafe budget, the likelihood multiplier has expectation at most one, so its probability of crossing $1/\delta$ is at most δ. A false family certificate requires passage of any one fixed unsafe budget. Thus each component may use allowance δ, without division by the number of budgets. This is ordinary intersection-union logic for an all-pass

assertion; it does not give individually selectable target conclusions.

On a margin-class table the mean log increment is at least d, and the upward overshoot is at most u. Bounded stopping and monotone convergence give expected proposals at most c per budget. For any item i, its probability of being acquired is at most one and at most its expected total number of proposals:

$$\Pr\{i \text{ is acquired}\} \le \min\{1, c \sum_{n \in \mathcal{K}} a_n(i)\}.$$

Summing over actual items gives B_c. Caching does not remove repeated proposals from the statistical stream. Reusing the same auxiliary coin on a repeated item would invalidate its iid argument. The proof charges distinct labels, not an equally small number of proposals or units of runtime.

For a factor-two-separated grid, finite winner-rank geometry gives $\sum_n w_n(r) \le 8e_r$. Under Theorem 1's parameter range, $c \le 17H/8$, so $B_c \le 17G$. Choosing the deterministic prefix instead whenever its bound is smaller proves $17F$. A finite proposal cap with a fully charged fallback gives $B_c + \eta$ for any prescribed $\eta > 0$; Appendix B accounts for that surcharge explicitly.

For an arbitrary grid, Appendix B groups budgets between factor-four representatives, inspects a shared prefix, and tests the remaining group risks through one empirical cumulative loss curve. Monotone likelihood ratios control a whole group without a target-count union bound. A geometric mixture inequality then bounds the sum of residual masses by the envelope tail, giving the more precise upper coefficient 61440, which is less than 2^{16}.

5. Why the same functional is unavoidable

The lower bound must place expensive inputs *inside* the entire safe margin class. A difficult table for one budget, or a different safe prior for each rank band, does not justify adding the resulting costs. Adaptive fixed-table query bounds are classical [8], as are rare-risk perturbations across geometric score bands [4]. The proof here combines two common-safe constructions to cover all finite regimes.

Choose representatives by starting at $a = \min \mathcal{K}$ and taking the next declared budget at least four times the preceding one. Their rank bands, from just beyond $m/(4a_j)$ through m/a_j, are disjoint and each has constant mass under its own representative. When many such bands contain sufficiently many items, a deterministic head removes the heavy coordinates. A product prior on the suffix is conditioned on the exact whole-family safe event. A maximal inequality for monotone rows controls the exceptional mass without a grid-size penalty.

Information is calculated under the *unconditioned* product laws using finite capped transcripts, with the exceptional mass charged explicitly. Each retained band requires order H expected acquisitions under the same conditional safe prior, so their costs add. This supplies the broad scale part of the lower bound.

When no broad scale remains and the smallest budget is below $m/4$, that budget dominates all larger budgets on a tail of constant mass. Centered fixed-count perturbations in disjoint tail blocks preserve the smallest-budget risk exactly at $\tau - \varepsilon$; domination keeps every null table safe for the whole grid. Changing one block's count gives a strictly unsafe alternative. The finite-population testing lemma in Appendix C controls depletion even under adaptive acquisition. Conditional-tail geometry compares the resulting block costs with F.

Appendix A also handles the high-budget endpoint using Appendix C's rankwise and hidden-subset bounds. The latter is needed for small prompt counts. The uniform-tail case covers budget one; finite endpoint checks cover small pools and partial supports. These arguments account for both clipping and deterministic truncation, rather than inserting a census cap into an asymptotic lower bound.

6. A subcensus regime and a finite limitation

Take one pool, $m = 2^d$, and every dyadic budget:

$$P = 1, \qquad \mathcal{K} = \{1, 2, 4, \ldots, m\}, \qquad \varepsilon^2 = \frac{\tau \log m}{m}.$$

Fix admissible τ, δ. For sufficiently large m the margin condition holds and $H = Lm/\log m$. The representative bands with $m/a_j \geq 32H$ number $\Theta(\log \log m)$, giving the lower rate. For the upper rate, $e_r \leq 2/r$ when $r \geq 2$ bounds the clipped sum by order $H[1 + \log(m/H)]$. The separated-grid construction therefore gives

$$C^*(\varepsilon) = \Theta\left(\frac{m \log \log m}{\log m}\right) = o(m).$$

Constants here may depend on the fixed τ, δ. For this sequence, the earlier blanket bound of order $\min\{m, H\sum_r e_r\}$ was only of census order because $\sum_r e_r = \Theta(\log m)$. The clipped analysis resolves a smaller matching order. This compares proved guarantees, not observed model performance or the actual cost on every individual safe table.

A separate exact finite analysis [1] concerns 32 prompts, 64 candidates, budgets $1, 2, 4, 8, 16, 32, 64$, threshold $1/10$, margin $1/160$, and error allowance $1/20$. Its current bounds are

$$1103.263793259 \leq C^*(1/160) \leq \frac{383040}{199}.$$

The upper endpoint is approximately 1924.824 labels. It certifies without observing a label with probability $9/199$; otherwise it runs a charged evidence branch whose error allowance is $1/200$ and whose uniform expected cost is at most 2016. Its accounting is

$$\frac{9}{199} + \frac{190}{199}\frac{1}{200} = \frac{1}{20}, \qquad \frac{190}{199}2016 = \frac{383040}{199}.$$

This is a permitted certification lottery under the stated objective. The default callback disables blind certification and has the separate 2016-label guarantee at error $1/200$. The mixed bound improves 1934.2 by about 9.376 labels, or 0.485%; it is not an exact optimum or a measured LLM saving. Here $F = 2048$, so the generic theorem does not improve this instance-specific certificate. The complete checkpoint proof and lower-certificate proof remain separate; they are not reproduced in the appendices of this manuscript.

7. Evidence, attribution, and scope

The saved all-grid checker covers 27,600 exact finite configurations, including all nonempty grids through pool size ten and specified larger grids. It checks the finite geometry, centered slices, mixture bounds, and likelihood identities. Its output reproduced byte for byte in a separate run. These are arithmetic and implementation checks; they do not replace the proofs' universal quantifiers or constitute independent human verification [1].

The rational callback reference implements the cached construction with exact integer sampling, conservative likelihood arithmetic, a finite proposal cap, and fully charged fallback. Under its stated conditions it retains a $17F + \eta$ bound. It has bounded rational input and resource limits, whereas the ideal theorem allows real losses. It does not implement the arbitrary-grid grouped construction. Mathematical admissibility for all real tables must not be inferred from that narrower program alone.

The relevant foundations are substantial. Weighted finite-population confidence sequences and use of side information appear in Shekhar et al. [9]; sequential stratified inference and careful sample accounting appear in Spertus et al. [10]. Fitas [3] already studies envelope designs and adaptive acquisition limits for best-of-N evaluation. Sarkar and Baichoo [4] develop curve certification, multiscale lower bounds, and deterministic-grade reuse. Their experiments and outputs differ from the present all-pass fixed-table problem; that difference requires a comparison of theorems and reductions, not dismissal of their results.

The older weighted-query comparison is only partially resolved. Wirth's 2010 primary talk [6] explicitly presents heavy-item inspection followed by residual sampling, including its confidence logarithm, and deterministic tail truncation for geometric weights. Thus those mechanisms are not new here. The final 2011 journal proof [11] was not obtained. Its precise finite-size conditions, expected-cost quantifiers, and possible reductions to the present safe-side family objective remain unchecked. The bounded source comparison does not establish scientific priority.

The theorem assumes correct fixed labels and known ranks. It does not validate a grader or a harm rubric, account for noisy repeated labels, adapt to an unknown margin with the same guarantee, or infer safety on new prompts. An omitted budget is outside the certificate. Computation may greatly exceed acquired-label cost. No real-model output or human annotation study supports the claimed rates. The artifact makes a specific mathematical statement inspectable; practical usefulness, independent correctness, and research ownership remain separate tests.

Complete proof appendices

These appendices consolidate the existing proofs; they make no claim of new scientific results or independently established priority. The fixed experiment, admissible auditors, exact validity and all-safe completion requirements, and margin class $\mathcal{Y}_\varepsilon$ are those in the main text. In particular, labels are fixed and correct, losses may be fractional, cost counts distinct acquisitions, and the supplied margin is a design input rather than evidence. The lower bounds permit empty-query certification lotteries and adaptive query choices.

All logarithms are natural. Throughout, unless an expressly broader statement is given, the standing parameter conditions are

$$0<\tau\le\tfrac{1}{8},\qquad 0<\varepsilon\le\tfrac{\tau}{16},\qquad 0<\delta\le\tfrac{1}{4}.$$

We retain the common notation $a=\min\mathcal{K}$, $L=\log(1/\delta)$, $H=\tau L/\varepsilon^2$, $e_r=\max_{n\in\mathcal{K}} w_n(r)$, $G=\sum_r \min\{P,He_r\}$, and $F=\min\{Pk_\varepsilon,G\}$. The tails are

$$q_k=q_a(k)=\sum_{r>k} w_a(r)=\frac{\binom{m-k}{a}}{\binom{m}{a}},\qquad Z_{\mathrm{tail}}(k)=\sum_{r>k} e_r.$$

Here W_n denotes the rank returned by the budget-n selector. The index k ranges from zero to m, and $k_\varepsilon=\min\{k:q_k\le\varepsilon\}$. Binomial coefficients for impossible choices are zero. The envelope is nonincreasing, and

$$G=\min_{0\le k\le m}\{Pk+HZ_{\mathrm{tail}}(k)\},$$

because increasing k by one changes the objective by $P-He_{k+1}$. These increments are nondecreasing, including equality cases.

Appendix letters prefix all retained section numbers and equation labels. For example, (A1), (B5), and (C7) identify equations in different appendices. Original setup equations that are not needed again are omitted, so label gaps are intentional. Appendix A starts at A.2 because its source's Section 1 is supplied by the main text and these standing conventions. Appendix C starts at C.2 for the same reason. The source correspondence and surgical changes are recorded in `source-map.md`.

Appendix A. Universal lower bound and dyadic corollary

A.2 Representative bands and the residual lower quantity

Choose the usual representatives: $a_1=\min\mathcal{K}$, then the smallest declared budget at least four times the preceding representative, until none remains. For representative a_j, put

$$I_j=\{\lfloor m/(4a_j)\rfloor+1,\dots,\lfloor m/a_j\rfloor\},\qquad S_j=Pm/a_j.$$

Let

$$J_{32}=\#\{j:S_j\ge 32H\}.$$

Proposition. Every finite nonempty declared best-of-n grid satisfies

$$C^*(\varepsilon)\ge HJ_{32}/4096. \tag{A1}$$

The same conditional prior supported entirely on the exact efficiency class serves every summand. No maximum-atom assumption on the original target rows and no minimum-budget restriction are needed.

The representative facts

Every declared budget in the group based at a_j lies in $[a_j, 4a_j)$. For $a \le n$,

$$w_n(r) \le \tfrac{n}{a} w_a(r).$$

On positive support this follows from

$$w_n(r) = \tfrac{n}{m} \prod_{u=1}^{r-1} (1 - \tfrac{n-1}{m-u}),$$

because every factor decreases with n; terminal zeros preserve the inequality. Thus each group envelope is at most $4w_{a_j}$.

The bands I_j are disjoint since $a_{j+1} \ge 4a_j$. Each has mass $c_j > 1/4$ under its own representative. To check the finite endpoints, write $a = a_j$, $h = \lfloor m/a \rfloor \ge 1$, and $\ell_0 = \lfloor m/(4a) \rfloor = \lfloor h/4 \rfloor$. A union bound gives $\Pr(W_a \le \ell_0) \le a\ell_0/m \le 1/4$, and the exact tail gives

$$\Pr(W_a > h) = \tfrac{\binom{m-h}{a}}{\binom{m}{a}} \le (1 - h/m)^a \le e^{-ah/m}, \qquad ah/m > h/(h+1).$$

If $h \le 3$, then $\ell_0 = 0$ and the band mass is greater than $1 - e^{-1/2} > 1/4$. If $h \ge 4$, it is greater than $3/4 - e^{-4/5} > 1/4$. Impossible binomial choices make the tail zero, so this includes $a = m$ and $m = 1$.

A.3 A maximal lemma for residual rows

Let $X_1, \dots, X_D$ be independent Bernoulli variables with common mean p. Fix $v \ge p$, $v > 0$, $0 < t \le 2v$, and $\rho > 0$. Consider any family of nonnegative decreasing rows b satisfying

$$b_1 \ge \cdots \ge b_D \ge 0, \qquad \sum_{i=1}^{D} b_i \le 1, \qquad b_1 \le \rho.$$

Then

$$\Pr\{\exists b : \sum_{i=1}^{D} b_i (X_i - p) > t\} \le \exp[-t^2/(4v\rho)]. \tag{A2}$$

Proof. Put $S_s = \sum_{i \le s} (X_i - p)$ and $\lambda = t/(2v) \le 1$. For $0 \le \lambda \le 1$,

$$\begin{aligned} \log \mathbb{E} e^{\lambda(X_i - p)} &= \log(1 + p(e^\lambda - 1)) - p\lambda \\ &\le p(e^\lambda - 1 - \lambda) \\ &\le p\lambda^2 \le \lambda t/2. \end{aligned} \tag{A2a}$$

Here $\log(1+z) \le z$ for $z \ge 0$, and $e^\lambda - 1 - \lambda \le \lambda^2$ on $[0,1]$; the latter follows by bounding the exponential series by

$$\lambda^2 \sum_{k \ge 2} 1/k! = (e-2)\lambda^2 \le \lambda^2.$$

The last inequality uses $p \le v$ and $v\lambda = t/2$. Independence therefore makes

$$M_s = \exp\{\lambda(S_s - st/2)\}, \qquad M_0 = 1,$$

a nonnegative supermartingale. Its classical finite-horizon maximal inequality [14] gives

$$\Pr\{\exists s \le D : S_s > \tfrac{t}{2}(s + 1/\rho)\} \le \exp[-t^2/(4v\rho)]. \tag{A2b}$$

Indeed, such a prefix forces $M_s > \exp(\lambda t/(2\rho))$. On the complementary event, append $b_{D+1} = 0$ and use summation by parts:

$$\begin{aligned} \sum_{i=1}^{D} b_i(X_i - p) &= \sum_{s=1}^{D} (b_s - b_{s+1}) S_s \\ &\le \tfrac{t}{2}\Big[\sum_{s=1}^{D} s(b_s - b_{s+1}) + \tfrac{1}{\rho}\sum_{s=1}^{D}(b_s - b_{s+1})\Big] \\ &= \tfrac{t}{2}\Big(\sum_{i=1}^{D} b_i + \tfrac{b_1}{\rho}\Big) \le t. \end{aligned} \tag{A2c}$$

The same prefix event works for every row, including terminal zero weights and noninteger $1/\rho$. Thus subprobability rows require no normalization and no target-cardinality union bound. QED

A.4 A common safe prior with a deterministic head

Put

$$k = \lfloor 8H/P \rfloor, \qquad p = \tau - 2\varepsilon.$$

If $k \ge m$, then $Pm \le 8H$, and hence $S_j \le Pm < 32H$ for every representative. In that case $J_{32} = 0$ and (A1) is immediate. Hence suppose $k < m$.

Under the unconditioned null prior Π, set every head loss at rank $r \le k$ to the fixed value p; set every suffix loss at $r > k$ independently to a Bernoulli(p) bit. This is a finite distribution over fixed correct-loss tables. The fractional head entries are allowed by the original model.

Every decreasing normalized rank row satisfies

$$(k+1)w_n(k+1) \le 1.$$

Therefore every suffix item row, in rank-major order, has total mass at most one and maximum entry at most

$$\rho = 1/[P(k+1)] < 1/(8H). \tag{A3}$$

The expected full risk remains p for every target: the deterministic head and random suffix both have mean p. Applying (A2) with $v = \tau$ and $t = \varepsilon$ proves, for

$$S = \{\max_{n \in \mathcal{K}} R_n \le \tau - \varepsilon\}, \qquad \eta_S = \Pi(S^c),$$

that

$$\eta_S \le \exp[-\varepsilon^2/(4\tau\rho)] \le e^{-2L} = \delta^2 \le \delta/4 \le 1/16. \tag{A4}$$

The conditional law $\Pi_S = \Pi(\cdot \mid S)$ is consequently well defined and is supported on the exact original efficiency class. Its labels are not claimed to be independent. Information calculations below use Π, not conditional-label kernels under Π_S.

A.5 Alternatives on the retained bands

For any retained representative with $S_j \geq 32H$,

$$\frac{m}{4a_j} = \frac{S_j}{4P} \geq \frac{8H}{P} \geq k.$$

As k is integral, $\lfloor m/(4a_j) \rfloor \geq k$, so the entire band I_j is in the random suffix. The band calculation in Section A.2 gives

$$c_j := \sum_{r \in I_j} w_{a_j}(r) > 1/4. \tag{A5}$$

All representative bands are disjoint, including the retained ones.

Let Q_j equal Π except that the independent label mean inside this one band is

$$q = p + 12\varepsilon = \tau + 10\varepsilon \leq 13\tau/8 < 1/4.$$

The full witness risk under Q_j has mean

$$\mu_j = p + 12\varepsilon c_j > \tau + \varepsilon. \tag{A6}$$

Only the suffix is random. Its weights have sum at most one and each is at most ρ; all its success probabilities are at most q. The ordinary Bernoulli lower-tail moment bound therefore gives, for $s \geq 0$,

$$\log \mathbb{E}_{Q_j} \exp[-s(R_{a_j} - \mu_j)] \leq s^2 q\rho/2.$$

Optimizing and using (A3), (A6), and $q \leq 2\tau$,

$$Q_j\{R_{a_j} \leq \tau\} \leq \exp[-\varepsilon^2/(2q\rho)] \leq \exp[-\varepsilon^2/(4\tau\rho)] \leq \delta/4. \tag{A7}$$

For every event E consisting only of certificate issuance, fixed-table global validity and (A7) imply

$$Q_j(E) \leq \delta + \delta/4 = 5\delta/4. \tag{A8}$$

Thus the alternative need not be conditioned; the mass of its possibly safe tables is charged explicitly.

A.6 Finite transcripts and the residual lower bound

Fix any admissible auditor. Let T_j be its distinct queries in band I_j before certification on safe tables, and set

$$x_j = \mathbb{E}_{\Pi_S} T_j, \qquad \kappa = 256\varepsilon^2/\tau, \qquad B = \lceil H/512 \rceil = \lceil L/(2\kappa) \rceil.$$

If any retained x_j is infinite, (A1) is immediate. Otherwise stop the transcript at the first of certification, the Bth distinct band query, or a finite transition horizon h. Certification simultaneous with the Bth band query counts as reaching the cap. Let E_h be certification by that horizon strictly before the band cap.

Under the unconditioned product priors Π and Q_j, every unseen band label has its original Bernoulli(p) or Bernoulli(q) law, even with adaptive query choices. Its divergence is at most

$$\mathrm{kl}(p, q) \leq \frac{(12\varepsilon)^2}{q(1-q)} \leq \frac{144\varepsilon^2}{(7\tau/8)(3/4)} < \kappa. \tag{A9}$$

Every head label is deterministic and identical under both priors. Every outside-band suffix label has an independent unchanged law. Choices, randomization, and stopping use identical conditional rules given observed histories. The finite-horizon chain rule charges only band labels, giving

$$\mathrm{KL}(\Pi^{(h)}, Q_j^{(h)}) \leq \kappa[(1 - \eta_S)x_j + \eta_S B]. \tag{A10}$$

On S the truncated band count is at most the final T_j; on its complement it is at most B. No cost promise outside S is used.

Suppose $x_j < B/8$. Almost-sure safe-table certification and Markov's inequality imply

$$\Pi\{\text{certify strictly before band cap}\} \geq (1-\eta_S)(1-x_j/B) > (15/16)(7/8) > 3/4.$$

For some sufficiently large finite horizon, $\Pi(E_h) \geq 3/4$. From (A8), binary data processing, and $L \geq \log 4$,

$$\mathrm{KL}(\Pi^{(h)}, Q_j^{(h)}) \geq \tfrac{3}{4}[L - \log(5/4)] - \log 2 \geq L/8. \tag{A11}$$

On the other hand, $\kappa \leq \tau \leq 1/8 \leq L/8$, so $\kappa B \leq L/2 + \kappa \leq 5L/8$. Equations (A4) and (A10) imply

$$\mathrm{KL}(\Pi^{(h)}, Q_j^{(h)}) < \kappa B(1/8 + 1/16) \leq 15L/128 < L/8,$$

contradicting (A11). Consequently

$$x_j \geq B/8 \geq H/4096. \tag{A12}$$

Only the certificate events use increasing-horizon limits; every KL calculation is at a finite horizon. Summing (A12) over the disjoint retained bands under the same safe conditional prior gives expected total cost at least $HJ_{32}/4096$. Some fixed table in that finite support has at least the average expected cost. Taking the minimax infimum proves (A1).

A.7 A centered common-safe prior on the domination tail

Write $a = \min \mathcal{K}$, and suppose $a < m/4$. Put

$$k_0 = \lfloor m/(a+1) \rfloor, m' = m - k_0,$$
$$q = \Pr(W_a > k_0), T = \{k_0 + 1, \ldots, m - a + 1\}.$$

The exact adjacent-budget ratio is

$$\frac{w_{n+1}(r)}{w_n(r)} = \frac{n+1}{n}\frac{m-r-n+1}{m-n}.$$

Where the weights are positive this ratio is at most one exactly when $(n+1)r \geq m+1$. Terminal zeros preserve the inequality. Thus

$$w_n(r) \leq w_a(r) \quad (n \geq a,\ r \in T). \tag{A13}$$

Every nonnegative table supported on T satisfies $R_n \leq R_a$ for all originally declared targets.

Also $q \geq 1/4$. Indeed, put $x = m/(a+1) \geq 2$. Every denominator in the rank-by-rank product for q exceeds $m - x = ax$, giving

$$q = \prod_{u=0}^{k_0-1} (1 - \tfrac{a}{m-u}) \geq (1 - 1/x)^{k_0} \geq (1 - 1/x)^x \geq 1/4. \tag{A14}$$

The final inequality follows because $(1 - 1/x)^x$ increases for $x \geq 2$.

Consider disjoint blocks of tail cells. A block has $N \geq 3$ cells with positive original item weights $a_i = w_a(r_i)/P$, and put

$$s = N \min_i a_i .$$

Centered tail block lemma. If every selected block has $s \geq 8\varepsilon$, then one common safe prior gives

$$C^*(\varepsilon) \geq 2^{-17} \sum_{\text{selected blocks}} \min(N, Hs^2). \tag{A15}$$

Here blocks need not have comparable maximum and minimum weights.

To prove this, set every loss outside $\mathcal{T}$ to zero and every unassigned tail cell to

$$p_0 = (\tau - \varepsilon)/q.$$

Equations (A14) and the parameter range give $15\tau/16 \leq p_0 \leq 1/2$. Independently in each block, choose a uniform k-subset, put $p = k/N$, and assign

$$k = \lfloor N\tau/8 \rfloor, \qquad y_i = p_0 + \frac{\min_l a_l}{2a_i}(Z_i - p). \tag{A16}$$

The coefficient lies in $(0, 1/2]$. Since $p \leq \tau/8$, the losses are nonnegative, and their maximum is at most $p_0 + 1/2 \leq 1$. Each block's weighted risk is exactly p_0 times its original target mass, because $\sum_i (Z_i - p) = 0$. Therefore every null table has $R_a = p_0 q = \tau - \varepsilon$, and (A13) puts it in the original efficiency class.

For one block at a time change its count to $k + d$, where

$$d = \lfloor 2\varepsilon N/s \rfloor + 1.$$

Retain the same affine label maps and unchanged independent outside law. The witness risk increases by $sd/(2N) > \varepsilon$, so every alternative is strictly unsafe. Moreover

$$k + d \leq N/64 + N/4 + 1 \leq N/2 \qquad (N \geq 5).$$

For $N = 3, 4$, one has $k = 0$ and respectively $d \leq 1, 2$, so the same count condition holds. Each label reveals its bit through a known invertible map. Thus the finite-population lemma in Section C.2 applies exactly, with no conditioning of slice placements, giving block cost at least

$$\tfrac{1}{1024} \min\{N, N(k+1)L/d^2\}.$$

Put $z = \varepsilon N/s$. If $z \geq 1$, then $d \leq 3z$ and $k + 1 > N\tau/8$, whence the information term is at least $Hs^2/72$. If $z < 1$, then $d \leq 2$ and the information term is at least $NL/4 \geq N/4$. In both cases the displayed cost is at least $\min\{N, Hs^2\}/(72 \cdot 1024)$, hence at least $2^{-17} \min\{N, Hs^2\}$. All blocks use the same independent product of exact-count null laws. Their expected distinct query counts add; taking a fixed table with at least the average cost proves (A15).

Application of the finite-population testing lemma

Section C.2 proves the exact-count lower bound used above, including its $k = 0$ case and independent, unchanged outside law. That lemma is stated for a common positive amplitude and zero labels. It also applies to a cellwise known invertible map $Z_i \mapsto b_i + c_i Z_i$ with $c_i > 0$: given the queried identity, the two observed outcomes are an invertible relabeling of the bit, so each conditional KL divergence is unchanged. This is a proof reduction; no actual label or profile promise is supplied to the auditor for free. Each coordinate in (A16) uses such a map.

A.8 Conditional-tail geometry, including block length three

For $a \geq 2$, the conditional tail is another exact best-of-a law:

$$w_a(k_0 + r) = q v_r, \qquad v_r = \frac{\binom{m'-r}{a-1}}{\binom{m'}{a}}. \tag{A17}$$

Since $a < m/4$ implies $m \ge 4a+1$, $m' \ge ma/(a+1) \ge 3a$. Put

$$\varepsilon' = \varepsilon/q \le 1/32, H' = Hq^2,$$
$$\ell = \lfloor m'/a \rfloor \ge 3, D' = m' - a + 1,$$
$$U = \lceil D'/2 \rceil, J' = \lfloor U/\ell \rfloor.$$

Here $J' \ge 1$, because $D' - (2\ell - 1) \ge (a-2)(\ell - 1) \ge 0$, and $U \le m'/2$. Use conditional-rank blocks

$$B_j = \{(j-1)\ell + 1, \ldots, j\ell\},$$
$$N = P\ell, \quad s'_j = \ell v_{j\ell} \quad (1 \le j \le J').$$

Their original-weight minimum masses are $s_j = q s'_j$. Define

$$R = \#\{j : s'_j \ge 8\varepsilon'\}, \quad A = \sum_{j=1}^{R} \min(N, H'(s'_j)^2). \tag{A18}$$

Let $Q'_t = \sum_{r>t} v_r$, $k'_{\varepsilon'} = \min\{t : Q'_t \le \varepsilon'\}$, and

$$G' = \sum_r \min(P, H' v_r), \quad F' = \min(P k'_{\varepsilon'}, G').$$

The needed pure-geometric statement is

$$R \ge 1, \quad F' \le 8192A. \tag{A19}$$

First block

If $\ell = 3$, then $a/m' > 1/4$, and the first two adjacent ratios of v are at least $2/3$ because $m' \ge 3a$. Thus $s'_1 = 3v_3 \ge 1/3$.

If $\ell \ge 4$, the exact hazard identity gives

$$s'_1 = \frac{\ell a}{m' - \ell + 1} Q'_{\ell - 1}.$$

Every factor in the a-factor product for $Q'_{\ell-1}$ exceeds $1 - 1/a$. Also $m' \le a\ell + a - 1$, so the prefactor is at least $a/(5a/4 - 1)$. For $a = 2, 3$ this gives lower bounds $1/3$ and $32/99$. For $a \ge 4$, use $(1 - 1/a)^{a-1} > 1/3$ to obtain $s'_1 > (a-1)/[3(5a/4 - 1)] \ge 1/4$. Consequently $s'_1 \ge 1/4 \ge 8\varepsilon'$, proving $R \ge 1$.

Pairing, comparability, and decay

Log-concavity of the positive best-of-a weights gives

$$v_{2r-1} \le v_r^2 / v_1.$$

Pairing ranks, and padding by zeros outside the positive support, yields

$$G' \le 2 \sum_{r=1}^{U} \min(P, H' v_r^2 / v_1). \tag{A20}$$

Within the first half of the pool every adjacent ratio is at least $1 - 2/\ell$. Since $(1 - 2/\ell)^{\ell} \ge 1/27$ for integers $\ell \ge 3$, each full block's weights are at most 27 times its endpoint weight. Further, $\ell v_1 \ge \ell/(\ell + 1) \ge 3/4$. Therefore a full block contributes at most

$$\ell \min(P, 729 H' v_{j\ell}^2 / v_1) \le 972 \min(N, H'(s'_j)^2)$$

to the sum in (A20). The remaining fewer than ℓ ranks before U contribute at most $(4/3)\min(N, H'(s'_{J'})^2)$. Thus, writing $A_{\mathrm{all}} = \sum_{j=1}^{J'} \min\{N, H'(s'_j)^2\}$,

$$G' \le 2048 A_{\mathrm{all}}. \tag{A21}$$

Consecutive block endpoints also satisfy

$$s'_{j+1} \le (8/11) s'_j. \tag{A22}$$

Indeed every adjacent ratio is at most $1-(a-1)/(m'-1)$, while $\ell(a-1)/(m'-1) \ge 3/8$, and $(1-x)^\ell \le 1/(1+\ell x)$.

Removing small blocks

If $H'(8\varepsilon')^2 \le N$, equation (A22) gives

$$\sum_{j>R} \min(N, H'(s'_j)^2) \le \frac{H'(8\varepsilon')^2}{1-(8/11)^2} < 3H'(8\varepsilon')^2 \le 3A.$$

Thus (A21) implies $F' \le G' \le 8192A$.

If $H'(8\varepsilon')^2 > N$, every retained block contributes N to A. When $R = J'$, the positive support has size $D' \le 2U < 2(J'+1)\ell \le 4J'\ell$, so $Pk'_{\varepsilon'} \le 4A$. When $R < J'$, the exact tail identity gives

$$Q'_{(R+1)\ell} \le (m'/a) v_{(R+1)\ell} < (32/3)\varepsilon'.$$

Each additional ℓ ranks shrink the tail by at most $(1-a/m')^\ell \le 4/7$. Once the positive support is exhausted the tail is zero, so this statement also covers blocks extending past that support. Since $(32/3)(4/7)^5 < 1$, $k'_{\varepsilon'} \le (R+6)\ell \le 7R\ell$; hence $Pk'_{\varepsilon'} \le 7A$. This proves (A19) in all cases.

All retained blocks have $N \ge 3$ and $s_j = q s'_j \ge 8\varepsilon$, so (A15) and (A18) give the original-grid lower bound

$$C^*(\varepsilon) \ge A/2^{17}. \tag{A23}$$

A.9 Deterministic accounting against the full clipped envelope

Let

$$F = \min(Pk_\varepsilon, G), \quad G = \sum_r \min(P, He_r), \quad e_r = \max_{n \in \mathcal{K}} w_n(r).$$

Continue to write J_{32} for the number of broad representatives in Section A.2. If $J_{32} \ge 1$, split the greedy groups into those first J_{32} groups and the remaining groups. The first part has envelope at most $4 \sum_{j \le J_{32}} w_{a_j}$, so its clipped contribution is at most $4HJ_{32}$.

If there are remaining groups, let b be their smallest budget. Then $Pm/b < 32H$. The pointwise domination calculation (A13), now using b, gives envelope w_b after rank $\lfloor m/(b+1) \rfloor$. Censusing that prefix and charging the remaining row mass bounds this part by

$$P\lfloor m/(b+1) \rfloor + H < 33H.$$

Clipping is subadditive, so

$$F \le G \le 37HJ_{32} \qquad (J_{32} \ge 1). \tag{A24}$$

If $J_{32}=0$, put $S_0=Pm/a<32H$, and define the smallest-budget singleton quantities

$$G_a=\sum_r \min(P,Hw_a(r)), \qquad F_a=\min(Pk_\varepsilon,G_a).$$

Since every $w_a(r)\le a/m=P/S_0$,

$$G_a\ge \min(S_0,H)\ge S_0/32. \tag{A25}$$

Pointwise domination after $k_0=\lfloor m/(a+1)\rfloor$ gives $G\le Pk_0+G_a\le S_0+G_a\le 33G_a$. Therefore

$$F\le 33F_a \qquad (J_{32}=0). \tag{A26}$$

For $2\le a<m/4$, use Sections A.7--A.8. Since $q>\varepsilon$, the exact conditional-tail identity implies

$$k_\varepsilon=k_0+k'_{\varepsilon'}.$$

Also clipping the original tail at Hq instead of Hq^2 costs at most a factor $1/q\le 4$, so

$$G_a\le Pk_0+4G', \qquad F_a\le Pk_0+4F'. \tag{A27}$$

The first term in A is at least $\min(P\ell,H'/16)$. Since $m'\ge m/2$, $\ell\ge 3m'/(4a)$, $q\ge 1/4$, and $S_0<32H$,

$$A\ge \min(3S_0/8,H/256)\ge S_0/8192. \tag{A28}$$

Using (A19), (A27), and $Pk_0\le S_0$,

$$F_a\le 8192A+4\cdot 8192A=40960A. \tag{A29}$$

Combining (A26) and (A29),

$$F\le 1{,}351{,}680A<2^{21}A. \tag{A30}$$

A.10 Complete universal lower bound and endpoint cases

Universal lower bound. Under exactly the original rare-risk parameter range and finite fractional-table objective, every nonempty declared grid satisfies

$$C^*(\varepsilon)\ge F/2^{38}. \tag{A31}$$

If $J_{32}\ge 1$, (A1) and (A24) give the stronger bound $C^*\ge F/(37\cdot 4096)\ge F/2^{18}$. If $J_{32}=0$ and $a\ge m/4$, the high-budget result proved in Sections C.2--C.5 gives $C^*\ge F/2^{21}$. If $J_{32}=0$ and $2\le a<m/4$, combine (A23) and (A30) to get (A31).

It remains to cover $J_{32}=0$, $a=1$, $m>4$. The domination tail has $m'=\lceil m/2\rceil\ge 3$, $q=m'/m\ge 1/2$, and uniform weights on its $N=Pm'$ cells. Use this whole tail as one block in (A15), for which $s=q$. Then

$$C^*(\varepsilon)\ge 2^{-17}\min(Pm',Hq^2)\ge 2^{-19}\min(Pm,H)\ge 2^{-19}F_a.$$

Equation (A26) gives $C^*\ge F/(33\cdot 2^{19})\ge F/2^{25}$, which is stronger than (A31). Cases with $a=1, m\le 4$ were already in the high-budget case. This exhausts every finite grid.

The constant is intentionally loose. The completion is structural: broad scales use one safe conditioned product prior; the complementary regime uses an exact common-safe prior supported on the smallest-budget domination tail. Neither construction trusts the supplied margin on an unknown table. Every alternative is handled under fixed-table global validity, and all cost lower bounds permit adaptive acquisition and arbitrary certification lotteries.

A.11 Completion by the upper construction

Appendix B establishes in the same experiment

$$C^*(\varepsilon) \le \min\{Pk_\varepsilon, 61440G\} \le 2^{16}F,$$

and, on a factor-two-separated grid, $C^*(\varepsilon) \le 17F$. Section B.6 supplies the arbitrary-grid construction, with the residual concentration and stopping calculations proved in Section C.6. Sections B.2--B.5 give the cached likelihood construction and its finite completion, stopping, and expected coordinate-cost arguments. Combining these upper bounds with Section A.10 proves the main theorem.

A.12 A dyadic matching-rate corollary

Fix $\tau \in (0, 1/8]$ and $\delta \in (0, 1/4]$. Along powers of two $m = 2^d \to \infty$, take

$$P = 1, \qquad \mathcal{K} = \{1, 2, 4, \ldots, m\}, \qquad \varepsilon^2 = \frac{\tau \log m}{m}.$$

For all sufficiently large m, the theorem's margin condition holds, and

$$H = \frac{Lm}{\log m}.$$

The factor-four representatives are $a_j = 4^{j-1}$ up to the largest such budget in the grid. For large m, the broad representative count is

$$J_{32} = \lfloor \log_4 (\frac{\log m}{32L}) \rfloor + 1 = \Theta(\log \log m).$$

Section A.6 gives $C^*(\varepsilon) \ge HJ_{32}/4096$, while (A24) gives $F \le 37HJ_{32}$. The original dyadic grid is factor-two separated, so Section B.5 yields

$$\frac{HJ_{32}}{4096} \le C^*(\varepsilon) \le 17F \le 629HJ_{32}.$$

Consequently

$$C^*(\varepsilon) = \Theta(\frac{m \log \log m}{\log m}).$$

The constants in this asymptotic statement may depend on the fixed τ, δ. This is a cost theorem for fixed tables, with expectations over audit randomization. It is not a measured model saving. It makes no claim that every safe table incurs the displayed worst-case cost.

Appendix B. Cached likelihood and arbitrary-grid upper constructions

B.1 Definitions and upper statements

Write $a = \min \mathcal{K}$, $D = Pm$, and

$$\begin{gathered} L = \log(1/\delta), H = \frac{\tau L}{\varepsilon^2}, \\ q_a(k) = \frac{\binom{m-k}{a}}{\binom{m}{a}}, k_\varepsilon = \min\{k \in \{0, \ldots, m\} : q_a(k) \le \varepsilon\}, \\ Z_{\text{tail}}(k) = \sum_{r>k} e_r, G = \sum_{r=1}^{m} \min\{P, He_r\}, \\ F = \min\{Pk_\varepsilon, G\}. \end{gathered} \tag{B3}$$

All logarithms are natural. Throughout the cost corollaries assume

$$0<\tau\le\tfrac{1}{8},\qquad 0<\varepsilon\le\tfrac{\tau}{16},\qquad 0<\delta\le\tfrac{1}{4}. \tag{B4}$$

Theorem 1: cached likelihood audit. Put $\theta=\tau-\varepsilon$ and

$$\begin{aligned} u&=\log\tfrac{1-\theta}{1-\tau},\\ d&=\mathrm{kl}(\mathrm{Ber}(\theta),\mathrm{Ber}(\tau)),\ c=\tfrac{L+u}{d},\\ B_c&=\sum_{r=1}^{m}\min\{P,c\sum_{n\in\mathcal{K}}w_n(r)\}. \end{aligned} \tag{B5}$$

There is an admissible auditor with margin-class expected distinct-label cost at most B_c. This statement needs only $0<\varepsilon<\tau<1$ and $0<\delta<1$. Under (B4), if successive declared budgets satisfy $n_{j+1}\ge 2n_j$, then

$$\begin{aligned} C^{*}(\varepsilon)&\le\min\{Pk_\varepsilon,B_c\}\\ &\le\min\{Pk_\varepsilon,17G\}\le 17F. \end{aligned} \tag{B6}$$

Singleton families satisfy the spacing condition. The original family $\{1,2,4,8,16,32,64\}$ does too. The guarantee is an expected distinct-label bound, not a bound of the same size on random proposals or computation.

Theorem 2: arbitrary-grid upper. Every nonempty declared grid under (B4) satisfies

$$\begin{aligned} C^{*}(\varepsilon)&\le\min\{Pk_\varepsilon,61440G\}\\ &\le 61440F\le 2^{16}F. \end{aligned} \tag{B7}$$

Appendix A proves the matching lower bound within universal constants. Neither upper statement supplies selectable simultaneous confidence intervals or authorizes certifying a chosen subset when the originally declared global assertion fails.

B.2 The cached likelihood auditor

Fix an order of the originally declared budgets. For each budget n, prepare an independent iid proposal stream with item law a_n. Maintain one cache shared across all budgets. On every proposal of item I_t:

1. Acquire its label only if it is absent from the cache. 2. Generate a fresh independent auxiliary coin $B_t\sim\mathrm{Bernoulli}(y_{I_t})$, including on repeated items. 3. Update

$$S_{n,t}=\sum_{s=1}^{t}[B_s\log\tfrac{\theta}{\tau}+(1-B_s)\log\tfrac{1-\theta}{1-\tau}]. \tag{B8}$$

Pass budget n statistically when $S_{n,t}\ge L$. Also, before requesting another proposal, check whether every positive-support label for this budget is cached. If so, evaluate R_n exactly, pass if $R_n\le\tau$, and otherwise terminate without the global certificate. Issue the single global certificate only after every budget has passed. An exact-support check after the last needed acquisition may be performed immediately.

All probability statements condition on the fixed table. Consequently the fresh pairs (I_t,B_t) in a budget's stream are iid, and its coin mean is $R_n(y)$. Caching changes the number of labels purchased; it does not discard duplicate proposals or change their law. Reusing one old auxiliary coin on repeated proposals would invalidate this iid argument.

Global validity

For a fixed unsafe budget with $q=R_n(y)>\tau$, the expected likelihood-ratio multiplier is

$$q\tfrac{\theta}{\tau}+(1-q)\tfrac{1-\theta}{1-\tau}=1+\tfrac{(\tau-q)(\tau-\theta)}{\tau(1-\tau)}\le 1. \tag{B9}$$

Thus $\exp(S_{n,t})$ is a nonnegative supermartingale starting at one. Its probability of ever reaching $1/\delta$ is at most δ. The same statement holds conditional on all previous budget streams, since the current proposal stream and fresh coin randomness are independent of them and the table is fixed.

On any unsafe table, select one fixed originally declared budget whose risk exceeds τ. A false global certificate requires that budget's statistical pass: exact evaluation cannot pass it. If an earlier budget terminates the audit, no global certificate occurs. Hence the global false-certificate probability is at most δ.

Each component uses error allowance δ, without division by $|\mathcal{K}|$. This is intersection--union error accounting for one all-pass decision; it makes no simultaneous-interval claim.

Completeness, including the boundary

Each positive-support item in a fixed budget's finite proposal distribution has strictly positive probability. An iid stream therefore visits every such item after a finite random number of proposals almost surely. Unless the budget has already passed statistically, its exact-support check then evaluates its risk. On a safe table that check passes, including when the risk equals τ. There are finitely many budgets, so every safe table receives the global certificate in finite time almost surely.

The exact-support rule acquires no labels for free. Every first visit to an uncached item is charged. It can require many proposals when some positive weights are tiny; the model charges distinct labels and leaves computation uncharged.

B.3 Expected proposals and distinct acquisitions

Let T_n^{stat} be the first crossing time of (B8) in the uncapped iid stream, without the exact-support stopping rule. For a margin-class table, $q = R_n(y) \le \theta$. The mean log increment is

$$\mu(q) = q\log\frac{\theta}{\tau} + (1-q)\log\frac{1-\theta}{1-\tau} \ge \mu(\theta) = d > 0. \tag{B10}$$

The increment on a zero coin is $u > 0$ and the increment on a one coin is negative. For every finite horizon h,

$$S_{n,\,T_n^{\mathrm{stat}}\wedge h} \le L + u. \tag{B11}$$

Before crossing, the sum is below L; its last upward increment is at most u. Summing predictable stopping indicators at this bounded horizon gives

$$d\,\mathbb{E}(T_n^{\mathrm{stat}}\wedge h) \le \mu(q)\mathbb{E}(T_n^{\mathrm{stat}}\wedge h) = \mathbb{E}S_{n,\,T_n^{\mathrm{stat}}\wedge h} \le L + u. \tag{B12}$$

Monotone convergence gives

$$\mathbb{E}T_n^{\mathrm{stat}} \le c. \tag{B13}$$

In particular this crossing is finite almost surely on the margin class. Only bounded-stopping identities were used before taking the expectation limit. The actual budget duration T_n stops no later than T_n^{stat}, because its exact-support rule may pass first.

Now fix an item $i = (p, r)$. Before proposal t, the event $T_n \ge t$ is determined by the past, while the next item has probability $a_n(i)$. Therefore

$$\begin{aligned}\mathbb{E}\sum_{t=1}^{T_n}\mathbf{1}\{I_{n,t} = i\} &= \sum_{t\ge 1}\Pr(T_n \ge t)a_n(i)\\ &= a_n(i)\mathbb{E}T_n \le c\,a_n(i).\end{aligned} \tag{B14}$$

The cache received from earlier budgets does not change this identity. Conditional on that earlier history, the next stream remains iid and the bound (B13) still applies. On the margin class every preceding budget passes almost surely, so each declared component is eventually reached.

Item i is acquired at most once. Its probability of ever being acquired is at most its expected total number of proposals across all streams, and also at most one. Consequently

$$\begin{aligned}\mathbb{E}_y T &\le \sum_i \min\{1, c \sum_{n\in\mathcal{K}} a_n(i)\} \\ &= \sum_{r=1}^{m} \min\{P, c \sum_{n\in\mathcal{K}} w_n(r)\} = B_c, \qquad y \in \mathcal{Y}_\varepsilon.\end{aligned} \tag{B15}$$

This is the finite clipping step. It uses an inclusion probability for each actual item, with no promise about an unknown support or loss-count profile.

An optional, explicitly charged census cap

The preceding auditor already has almost-sure finite completion. To obtain a deterministic bound on the number of proposals, fix any $\eta > 0$, put $J = |\mathcal{K}|$, and cap each budget at

$$M = \lceil \tfrac{cJD}{\eta} \rceil \tag{B16}$$

proposals. Check for statistical or exact passage on the last proposal before invoking the cap. At the first unpassed cap, acquire every remaining relevant label and evaluate the global assertion exactly. This auditor uses at most JM proposals and at most D distinct labels, and still has the stated global validity and completeness.

On the margin class, a fallback implies that some corresponding uncapped statistical crossing time exceeds M. By (B13), Markov's inequality, and a union bound,

$$\Pr(\text{fallback}) \le \tfrac{Jc}{M} \le \tfrac{\eta}{D}. \tag{B17}$$

Ordinary acquisitions before fallback still obey the itemwise bound in (B15). The census adds at most D labels, so this capped version has

$$\sup_{y\in\mathcal{Y}_\varepsilon} \mathbb{E}_y T \le B_c + \eta. \tag{B18}$$

A forced census after D proposals has no such uncharged itemwise guarantee: it may acquire a rare item that none of the statistical streams proposed. The coupon-completion auditor attains (B15) directly; the optional deterministic proposal cap pays the explicit extra term (B18).

B.4 A pointwise geometric-mixture inequality

Lemma. For any increasing factor-four-separated sequence of budgets $a_{j+1} \ge 4a_j$,

$$\sum_j w_{a_j}(r) \le \tfrac{14}{3} \max_j w_{a_j}(r) \tag{B19}$$

at every rank. For a factor-two-separated sequence, the corresponding bound with constant 8 holds.

Proof. Fix a rank. On the interval where all factors are positive, extend the integer-budget formula by

$$g(x) = \tfrac{1}{m} \prod_{u=1}^{r-1} \tfrac{m-u-x+1}{m-u}. \tag{B20}$$

At an integer budget with positive weight, $w_n(r) = ng(n)$. The function g is nonincreasing and log-concave, and $g(1) = 1/m$. For rank one the product is empty. All terminal zero weights can be discarded.

Call a positive representative low if $g(a_j) \geq g(1)/2$, and high otherwise. The low representatives form an initial segment. If a is its last member, factor-four spacing gives

$$\sum_{j:\,\text{low}} w_{a_j}(r) \leq g(1) \sum_{j:\,\text{low}} a_j \leq \tfrac{4}{3} g(1) a \leq \tfrac{8}{3} w_a(r). \tag{B21}$$

For consecutive positive high representatives $a < b$, necessarily $a > 1$. Concavity of $\log g$ on the three points $1, a, b$ gives

$$\frac{g(b)}{g(a)} \leq \left(\frac{g(a)}{g(1)}\right)^{(b-a)/(a-1)} \leq 2^{-(b/a-1)}. \tag{B22}$$

Here $(b-a)/(a-1) \geq b/a - 1$, and the base in the middle is below one half. For $x = b/a \geq 4$,

$$\frac{w_b(r)}{w_a(r)} \leq x 2^{-(x-1)} \leq \tfrac{1}{2}. \tag{B23}$$

The last function decreases on $[4, \infty)$. Hence the high weights sum to at most twice their first weight. Adding this to (B21) gives (B19), including empty low or high segments.

For factor-two spacing, the geometric sum in (B21) is at most $2a$, giving a low contribution at most four times the largest weight. Split the high sequence into its odd and even positions. Each subsequence is factor-four separated; (B22)--(B23) bound each sum by twice its first weight. Their combined contribution is at most four times the largest weight. The total is at most eight times that weight. QED

This is a pointwise statement. In particular, for any factor-four-separated representatives contained in the declared family and any prefix k,

$$\sum_{j} \sum_{r > k} w_{a_j}(r) \leq \tfrac{14}{3} Z_{\text{tail}}(k) \leq 5 Z_{\text{tail}}(k). \tag{B24}$$

No minimum-budget fraction or rank-decay assumption is used.

B.5 The separated-grid constant and deterministic alternative

For $f(x) = \mathrm{kl}(\mathrm{Ber}(x), \mathrm{Ber}(\tau))$, we have $f(\tau) = f'(\tau) = 0$ and

$$f''(x) = \frac{1}{x(1-x)} \geq \frac{1}{\tau} \qquad (\theta \leq x \leq \tau).$$

Integrating twice gives

$$d \geq \frac{\varepsilon^2}{2\tau}. \tag{B25}$$

Under (B4),

$$u = \log\left(1 + \frac{\varepsilon}{1-\tau}\right) \leq \frac{\varepsilon}{1-\tau} \leq \frac{1}{112} < \frac{L}{16}. \tag{B26}$$

The last inequality follows from $L \geq \log 4 > 1$. Thus

$$c \leq \tfrac{17}{8} H. \tag{B27}$$

The factor-two part of the mixture lemma and (B15) now imply

$$B_c \leq \sum_r \min\{P, 8ce_r\} \leq \sum_r \min\{P, 17He_r\} \leq 17G. \tag{B28}$$

For completeness, the deterministic alternative in (B6) queries every prompt position in ranks $1, \dots, k_\varepsilon$. With the resulting known prefix contribution b_n, use

$$U_n = b_n + q_a(k_\varepsilon). \tag{B29}$$

Winning-rank tails decrease with the budget, so U_n bounds the actual R_n for every declared n. On $\mathcal{Y}_\varepsilon$, $U_n \le R_n + q_a(k_\varepsilon) \le \tau$. If the deterministic check fails on another table, census the remaining positive-support items and evaluate the family exactly. This is globally zero-error, all-safe complete, and costs at most Pk_ε throughout the margin class.

Choose between this deterministic rule and the likelihood auditor using their design bounds, before any labels are observed. This proves (B6). The optional capped likelihood version has the additional η from (B18); choosing $\eta = F$ and combining with the deterministic rule gives the explicit bound $18F$ with finitely bounded proposal counts. The uncapped coupon-completion version gives $17F$ as stated.

B.6 An arbitrary-grid prefix and group construction

Take the existing greedy representatives: start at $a_1 = \min \mathcal{K}$, and after a_j choose the smallest declared budget at least $4a_j$, if one exists. Assign every declared budget to its preceding representative. Thus each group has

$$a_j \le n < 4a_j, \tag{B30}$$

and the representatives are factor-four separated. No undeclared budget is added to the certificate.

Fix a prefix $k \in \{0, \dots, m\}$ and acquire its Pk labels once. Let

$$b_n = \sum_p \sum_{r \le k} a_n(p, r) y_{p,r}, \qquad q_j = \sum_{r > k} w_{a_j}(r). \tag{B31}$$

If $q_j \le \varepsilon$, use the deterministic bounds $b_n + q_j$ for this group. They pass on every margin-class table. If they fail on another table, census the remaining group support and decide that group exactly. A group with $q_j = 0$ is already determined by the prefix.

For a group with $q_j > \varepsilon$, use the residual test proved in Section C.6. That argument only uses $a_j \le n \le 4a_j$ within the group, not the high-budget theorem's assumption $a_j \ge m/4$. The following records its parameters and the changes needed here.

Draw iid tail proposals with item probability $w_{a_j}(r)/(Pq_j)$, acquire each uncached label, and use fresh auxiliary coins. On the representative's support put $h_n(r) = w_n(r)/w_{a_j}(r)$, extended by zero outside it. The rank-product identity gives $0 \le h_n \le 4$, and h_n is nonincreasing in rank. The tail-risk estimates are

$$\widehat{R}^{\mathrm{tail}}_{n,t} = \frac{q_j}{t} \sum_{s=1}^{t} h_n(\mathrm{rank}(I_s)) B_s. \tag{B32}$$

The residual-concentration proof in Section C.6, combining binomial Bernstein, conditional DKW, and summation by parts, gives

$$\Pr\{ \max_{n \text{ in group}} |\widehat{R}^{\mathrm{tail}}_{n,t} - R^{\mathrm{tail}}_n| > 4 r_j(t, x) \} \le 4e^{-x},$$
$$r_j(t, x) = 4\sqrt{\frac{q_j \tau x}{t}} + \frac{2 q_j x}{t}. \tag{B33}$$

when the representative's actual tail risk μ_j is at most 2τ. This is one event for the entire group, without a union bound over its targets.

At stages $\ell = 1, 2, \ldots$, take

$$\alpha_\ell = \delta 2^{-\ell}, x_\ell = L + (\ell + 2)\log 2,$$
$$A_j = \frac{4096 q_j \tau}{\varepsilon^2}, t_{j,\ell} = \lceil A_j x_\ell \rceil. \tag{B34}$$

Pass the group if every member satisfies

$$b_n + \widehat{R}^{\text{tail}}_{n, t_{j,\ell}} + 4 r_j(t_{j,\ell}, x_\ell) \le \tau. \tag{B35}$$

For an unsafe group with $\mu_j \le 2\tau$, passage requires failure of (B33), of probability at most α_ℓ at that stage. If $\mu_j > 2\tau$, its declared representative is unsafe and passage forces its empirical tail risk below $\mu_j/2$. The same Bernoulli lower-tail bound as in Section C.6 gives probability at most

$$\exp[-t_{j,\ell}\mu_j/(8q_j)] \le \exp[-t_{j,\ell}\tau/(4q_j)] \le e^{-x_\ell} \le \alpha_\ell. \tag{B36}$$

Thus each unsafe group passes statistically with probability at most $\sum_\ell \alpha_\ell = \delta$.

On a margin-class table, $\mu_j \le \tau$ and the radius in (B34) is less than $\varepsilon/8$. On the event (B33), every left side in (B35) is therefore at most $\tau - \varepsilon + 8 r_j < \tau$. If U_j is the number of uncapped proposals until group passage, the stage-tail calculation in (C47) gives

$$\mathbb{E} U_j \le t_{j,1} + \sum_{\ell \ge 1} (t_{j,\ell+1} - t_{j,\ell}) \alpha_\ell \le 12288 H q_j. \tag{B37}$$

For this grouped construction, explicitly cap proposals at the number D_j of positive-support group items after the prefix. If no passage has occurred, query all remaining unknown group-support items and evaluate its risks exactly. Its distinct-label cost beyond the prefix is at most $\min\{U_j, D_j\}$ on the margin class: when $U_j < D_j$ it uses at most U_j labels; otherwise its entire support census uses at most D_j. This cap is valid for (B37), which charges total proposal expectation. It is not used to assert the itemwise clipping bound (B15).

Run all groups with one shared label cache, independent proposal streams, and error allowance δ for each group. Issue the global certificate only when every group passes. On an unsafe table, choose a fixed unsafe group; its error bound alone controls global false certification. All exact fallbacks preserve validity, and the finite caps give all-safe completeness. On the margin class, the groups with $q_j \le \varepsilon$ require no fallback, so they add no uncharged cost.

Using (B24) and (B37),

$$\begin{aligned} \sup_{y \in \mathcal{Y}_\varepsilon} \mathbb{E}_y T &\le Pk + 12288 H \sum_j q_j \\ &\le Pk + 61440 H Z_{\text{tail}}(k) \\ &\le 61440 \{ Pk + H Z_{\text{tail}}(k) \}. \end{aligned} \tag{B38}$$

The envelope e_r is nonincreasing because every target row is. Incrementing k in the last braces changes its value by $P - H e_{k+1}$, so

$$\min_{0 \le k \le m} \{ Pk + H Z_{\text{tail}}(k) \} = \sum_r \min\{P, H e_r\} = G. \tag{B39}$$

Optimize this fixed prefix and also consider the deterministic k_ε rule. This proves (B7). The extension from the high-budget proof in Section C.6 consists of partitioning the originally declared targets, sharing their prefix and label cache, giving each group the global decision's error allowance, and bounding the sum of residual masses by (B24).

Appendix C. Finite-population, rankwise, and residual dependencies

Sections C.2--C.4 require the standing parameter conditions but no minimum-budget restriction. Sections C.5--C.6 give the high-budget specialization with $a \geq m/4$, establishing

$$\frac{F}{2^{21}} \leq C^{*}(\varepsilon) \leq 12288F. \tag{C6}$$

For later references, repeat the prefix identity

$$G = \min_{0 \leq k \leq m} \{Pk + HZ_{\text{tail}}(k)\}. \tag{C4}$$

The residual proof in C.6 itself uses only that every budget tested with representative a obeys $a \leq n \leq 4a$. Appendix B applies that proof separately to its groups; it does not impose $a \geq m/4$ on the original grid. Thus these sections supply every project-specific lemma used in Appendices A and B.

C.2 An exact finite-population testing lemma

The following lemma concerns distributions over fixed oracle tables, not a noisy-label model.

Lemma. A block has N cells. Under a safe prior, exactly k cells have the same positive loss amplitude $\gamma \leq 1$, uniformly permuted, and all other block cells have loss zero. Under an alternative, exactly $k+d$ cells have amplitude γ, uniformly permuted. Assume

$$k \geq 0, \qquad d \geq 1, \qquad k + d \leq N/2. \tag{C7}$$

The outside table is independent of this block under both priors and has the same distribution. Every table in the safe prior's support belongs to $\mathcal{Y}_{\varepsilon}$, and every alternative table is unsafe. If χ is an admissible auditor's expected number of distinct block queries under the safe prior, then

$$X \geq \frac{1}{1024} \min\{N, \frac{N(k+1)L}{d^2}\}. \tag{C8}$$

Identical outside marginal laws without the stated independence would not suffice for the proof.

Proof. Write $R = \min\{N, N(k+1)L/d^2\}$. The probability of certifying without querying the block is the same under the two priors. Validity and safe-table completeness therefore imply $X \geq 1 - \delta \geq 3/4$. This proves (C8) when $R < 256$.

Otherwise set $b = \lfloor R/128 \rfloor$, so $b \geq R/256$ and $b \leq N/128$. Stop the transcript at certification, at the bth distinct block query, or at an additional finite transition horizon. A certificate simultaneous with the bth block query counts as reaching the cap. After $t < b$ distinct block queries with S_t positive labels, the next-label probabilities under the two priors are

$$u = \frac{k - S_t}{N - t}, \qquad v = \frac{k + d - S_t}{N - t}. \tag{C9}$$

Uniformity of the remaining placements holds under adaptive choices of cell identities and interleaving with the independent outside table. For every history of positive safe-prior probability, $0 < v < 1$ and

$$\begin{aligned} \mathrm{kl}(\mathrm{Ber}(u), \mathrm{Ber}(v)) &\leq \frac{(u-v)^2}{v(1-v)} \\ &= \frac{d^2}{(k+d-S_t)(N-t-k-d+S_t)} \\ &\leq \frac{4d^2}{N(k+d-S_t)}. \end{aligned} \tag{C10}$$

To control the expectation even on histories that reveal many positives, complete the block's success sequence after stopping to a uniform random permutation of k ones and $N-k$ zeros. The same sequential construction is valid for adaptive unseen-cell choices: each next unseen cell has the uniform remaining-success probability, and an unused suffix can be filled after the auditor stops. Thus S_t in this completed sequence has the hypergeometric distribution. Direct summation gives

$$\begin{aligned}\mathbb{E}\frac{1}{k-S_t+1} &= \frac{1}{\binom{N}{t}}\sum_s \frac{\binom{k}{s}\binom{N-k}{t-s}}{k-s+1} \\ &= \frac{1}{(k+1)\binom{N}{t}}\sum_s \binom{k+1}{s}\binom{N-k}{t-s} \\ &\le \frac{\binom{N+1}{t}}{(k+1)\binom{N}{t}} = \frac{N+1}{(k+1)(N+1-t)} \le \frac{2}{k+1}.\end{aligned} \tag{C11}$$

The sum is over the original feasible values of s. Extending that range only adds nonnegative terms; the inequality is Vandermonde's identity. Since $d \ge 1$, (C11) also bounds the reciprocal with denominator $k+d-S_t$.

Query choices, internal randomization, and decisions follow the same conditional rule given a transcript. Outside labels have the same conditional kernels, by independence and the unchanged outside law. The finite-transcript KL chain rule therefore charges only block labels. Replace each stopping indicator in the expected sum of (C10) by one and use the completed permutation sequence to obtain

$$\mathrm{KL}(\Pi^{(h)}, Q^{(h)}) \le \frac{8d^2 b}{N(k+1)} \le \frac{L}{16}. \tag{C12}$$

If $X < b/4$, Markov's inequality and almost-sure certification under the safe prior give probability greater than $3/4$ of certification strictly before the block cap. At a sufficiently large finite transition horizon h, its probability is at least $3/4$. Under the alternative it is at most δ. Binary data processing and $\mathrm{kl}(u, v) \ge u\log(1/v) - \log 2$ then give

$$\mathrm{KL}(\Pi^{(h)}, Q^{(h)}) \ge \frac{3L}{4} - \log 2 \ge \frac{L}{4}, \tag{C13}$$

contradicting (C12). Thus $X \ge b/4 \ge R/1024$. Only certificate-event probabilities take an increasing-horizon limit; every KL calculation is at a finite horizon. QED

C.3 Hidden subsets on a common fractional safe table

Let $p_0 = \tau - \varepsilon$ and take the single fixed table $y_{p,r} = p_0$. Every declared risk equals $\tau - \varepsilon$.

Lemma. For any rank with $e_r \ge 16\varepsilon$, an admissible auditor's expected number of queries at that rank on this table is at least

$$\frac{1}{1024}\min\{P, \frac{Le_r}{\varepsilon}\}. \tag{C14}$$

Proof. Choose a declared witness with weight $w = e_r$, put $h = 1 - p_0 \ge 7/8$, and let

$$s = \lfloor \frac{\varepsilon P}{hw} \rfloor + 1. \tag{C15}$$

Change any s cells in this rank from p_0 to 1, leaving all others unchanged. The witness risk becomes $p_0 + hws/P > \tau$. Also $1 \le s \le P$, because $\varepsilon/(hw) \le 1/14$; the case $P = 1$ gives $s = 1$ directly.

Let T be the rank-query count on the constant safe table. Average the alternative over all s-subsets. Couple the auditor's randomness between the two tables. If none of the changed cells is queried before the safe table's certificate, that entire transcript is unchanged and the alternative also certifies. Therefore

$$\mathbb{E}f(T) \le \delta, \qquad f(t) = \frac{\binom{P-t}{s}}{\binom{P}{s}}, \tag{C16}$$

where $f(t) = 0$ if $P - t < s$. In particular, $\mathbb{E}T \ge 1 - \delta \ge 3/4$.

Set $R_0 = \min\{P, PL/s\}$. If $R_0 < 64$, this already implies $\mathbb{E}T \geq R_0/256$. Otherwise set $b = \lfloor R_0/32 \rfloor$. Then $P \geq 64$, $b \geq R_0/64$, $b \leq P/32$, and $s \leq P/14 + 1 < P/2$. The product formula for $\jmath$ and $\log(1-u) \geq -2u$ for $0 \leq u \leq 1/2$ give

$$f(b) = \prod_{j=0}^{s-1} (1 - \tfrac{b}{P-j}) \geq \exp(-\tfrac{4sb}{P}) \geq e^{-L/8}. \tag{C17}$$

If $\mathbb{E}T < b/4$, then $\Pr(T < b) > 3/4$. Since $\jmath$ is nonincreasing, (C16)--(C17) would imply $\delta \geq \mathbb{E}f(T) > (3/4)\delta^{1/8} > \delta$. Thus $\mathbb{E}T \geq b/4 \geq R_0/256$ in this case as well.

Finally $s \leq 1 + (8/7)\varepsilon P/w$ implies

$$R_0 \geq \tfrac{7}{15} \min\{P, \tfrac{Lw}{\varepsilon}\}. \tag{C18}$$

For $\varepsilon P/w \geq 1$, use $s \leq (15/7)\varepsilon P/w$; for $\varepsilon P/w < 1$, use $s < 15/7$ and $L \geq \log 4 > 1$. Equations (C17)--(C18), with $7/(15 \cdot 256) > 1/1024$, prove (C14). QED

C.4 A rankwise lower bound for every declared grid

This section does not assume $a \geq m/4$.

Proposition. Under the standing parameter conditions, every nonempty declared grid satisfies

$$C^*(\varepsilon) \geq \tfrac{A_{16}}{32768}, \qquad A_{16} = \sum_{r\,:\,e_r \geq 16\varepsilon} \min\{P, He_r^2\}. \tag{C19}$$

Proof. First suppose $P\tau \geq 4$. Set

$$p_0 = \tau - \varepsilon, \qquad k = \lceil Pp_0 \rceil, \qquad \gamma = \tfrac{Pp_0}{k}. \tag{C20}$$

Independently at every rank, uniformly choose exactly k of the P prompt positions to have loss γ, and give all others loss zero. Every table in this common prior has rank-average loss exactly p_0 at every rank. Hence every declared risk is exactly $\tau - \varepsilon$; no conditioning or exceptional safe-table mass is involved. Moreover

$$P \geq 32, \qquad \gamma \geq \tfrac{3}{4}, \qquad k \leq \tfrac{P}{8} + 1 \leq \tfrac{5P}{32}. \tag{C21}$$

For a rank with $w = e_r \geq 16\varepsilon$, choose a declared budget attaining that envelope weight. Change only this rank's count to $k + d$, where

$$d = \lceil \tfrac{2\varepsilon P}{\gamma w} \rceil. \tag{C22}$$

Every table under this alternative has witness risk at least $p_0 + \gamma w d/P \geq \tau + \varepsilon > \tau$. Since $d \leq P/6 + 1$ and $P \geq 32$, (C21) gives $k + d < P/2$. The outside ranks are independent and unchanged, so the finite-population lemma applies.

Write $z = \varepsilon P/w$. If $z \geq 1$, then $d < 4z$ and

$$\tfrac{P(k+1)L}{d^2} \geq \tfrac{15}{256} Hw^2. \tag{C23}$$

If $z < 1$, then $d \leq 3$ and

$$\tfrac{P(k+1)L}{d^2} \geq \tfrac{5}{48} \tau P^2 L \geq \tfrac{5}{12} LP > \tfrac{P}{2}. \tag{C24}$$

In both cases $\min\{P, P(k+1)L/d^2\} \geq \min\{P, Hw^2\}/32$. Equation (C8) therefore gives expected rank cost at least $\min\{P, Hw^2\}/32768$. These rank counts add under the same safe prior. Some fixed table in its finite support has cost at least the average, proving (C19) when $P\tau \geq 4$.

When $P\tau < 4$, use the one constant safe table of Section C.3 and put

$$B_{16} = \sum_{r:\, e_r \ge 16\varepsilon} \min\{P, \tfrac{Le_r}{\varepsilon}\}. \tag{C25}$$

For $z = Le_r/\varepsilon \ge P$, the corresponding A_{16} term is at most the B_{16} term. If $z < P$, then

$$He_r^2 = \tfrac{\tau}{L} z^2 \le \tfrac{P\tau}{L} z < 3z. \tag{C26}$$

Thus $A_{16} \le 3B_{16}$. The rank bounds (C14) add on the same fixed table, giving $C^*(\varepsilon) \ge B_{16}/1024 \ge A_{16}/3072$. This is stronger than (C19) and completes the proposition. QED

C.5 High-budget envelope geometry

Assume now $a \ge m/4$ and write $c = 1/4$. Each individual positive row is log-concave, because

$$\tfrac{w_n(r+1)}{w_n(r)} = \tfrac{m-r-n+1}{m-r} \tag{C27}$$

is nonincreasing in r. Each row also has $w_n(1) = n/m \ge c$. Multiplying consecutive ratio inequalities, or using concavity of $\log w_n$, gives

$$w_n(2r-1) \le \tfrac{w_n(r)^2}{w_n(1)} \le \tfrac{w_n(r)^2}{c}. \tag{C28}$$

Extend rows by zero after their support; if $w_n(r) = 0$, the left side is also zero. Taking maxima over n in (C28) gives

$$e_{2r-1} \le \tfrac{e_r^2}{c}. \tag{C29}$$

This argument does not assume that the maximum of the log-concave rows is itself log-concave. Since the envelope is decreasing, grouping its terms in pairs and using (C29) yields

$$G \le \tfrac{2}{c} \sum_r \min\{P, He_r^2\}. \tag{C30}$$

If $m \le 4$, every positive best-of-n weight is at least $1/6$. Consequently every positive envelope weight exceeds $16\varepsilon \le 1/8$. The sum in (C30) is then A_{16} and $F \le 8A_{16}$.

For the remaining argument take $m \ge 5$. Every declared n is at least 2 and $m \le 4n$, so

$$\tfrac{n-1}{m-1} \ge \tfrac{n-1}{4n-1} \ge \tfrac{1}{7}. \tag{C31}$$

Thus every row, and then its envelope, satisfies

$$e_{r+1} \le \tfrac{6}{7} e_r. \tag{C32}$$

This weight-decay factor differs from the tail-decay factor below. Let $R = \#\{r : e_r \ge 16\varepsilon\}$. Since $e_1 \ge 1/4 > 16\varepsilon$, $R \ge 1$. For the smallest declared budget, the exact hazard and tail identities are

$$w_a(k+1) = \tfrac{a}{m-k} q_k, \qquad q_{k+1} = q_k(1 - \tfrac{a}{m-k}) \le \tfrac{3}{4} q_k \tag{C33}$$

whenever $q_k > 0$. If $q_R = 0$, already $k_\varepsilon \le R$. Otherwise $R < m$, $w_a(R+1) \le e_{R+1} < 16\varepsilon$, and (C33) gives

$$q_R < 16\varepsilon/c = 64\varepsilon.$$

Consequently,

$$\lceil \tfrac{\log 64}{\log(4/3)} \rceil = 15, \qquad k_\varepsilon \le R + 15 \le 16R. \tag{C34}$$

If $H(16\varepsilon)^2 > P$, each of the R terms in A_{16} equals P. It follows that

$$F \le Pk_\varepsilon \le 16A_{16}. \tag{C35}$$

If $H(16\varepsilon)^2 \le P$, (C32) instead gives

$$\begin{aligned}\sum_{r>R} \min\{P, He_r^2\} &\le \frac{H(16\varepsilon)^2}{1-(6/7)^2} \\ &\le \frac{H(16\varepsilon)^2}{c} \le \frac{A_{16}}{c}.\end{aligned} \tag{C36}$$

The last inequality uses $R \ge 1$ and the fact that an included term is at least $H(16\varepsilon)^2$. Combining (C30) and (C36) gives $F \le G \le 40A_{16}$. Together with the other cases, this proves

$$F \le 40A_{16}, \qquad C^*(\varepsilon) \ge \frac{F}{40 \cdot 32768} \ge \frac{F}{2^{21}}. \tag{C37}$$

C.6 A globally valid prefix-and-residual upper bound

First, querying every prompt position in the first k_ε ranks gives known prefix risks

$$b_n = \sum_p \sum_{r \le k_\varepsilon} a_n(p, r) y_{p,r}.$$

The tail probability is nonincreasing in the budget n, so $b_n + q_{k_\varepsilon}$ is a deterministic upper bound for every target risk. On a margin-class table it is at most $R_n(y) + q_{k_\varepsilon} \le \tau$. Certify if all these upper bounds pass; otherwise census the remaining positive-support cells and evaluate the risks exactly. This is globally valid and all-safe complete, and its cost on $\mathcal{Y}_\varepsilon$ is at most Pk_ε.

For the statistical part, fix any prefix k, query its Pk cells, and write b_n for its known risk contributions. Put $q = q_k$. If $q \le \varepsilon$, the same deterministic rule already gives the required margin-class cost Pk. Hence consider $q > \varepsilon$.

Since $a \le n \le m \le 4a$, the likelihood ratios

$$h_n(r) = \frac{w_n(r)}{w_a(r)} \tag{C38}$$

on the positive support of w_a satisfy $0 \le h_n(r) \le n/a \le 4$ and are nonincreasing in r. Indeed, the product formula

$$w_n(r) = \frac{n}{m} \prod_{u=1}^{r-1} \left(1 - \frac{n-1}{m-u}\right)$$

gives the bound, and on adjacent positive weights $h_n(r+1)/h_n(r) = (m-r-n+1)/(m-r-a+1) \le 1$. Terminal zeros preserve monotonicity. This permits the equality endpoint $n/a = 4$; no strict inequality is needed. The support of w_n is contained in that of w_a, and ratios are extended by zero outside the latter.

Draw iid tail items with probability $w_a(r)/(Pq)$ for each (p, r) with $r > k$. Query an item's fixed loss only if it is not already cached, and generate a fresh auxiliary coin $B \sim \mathrm{Bernoulli}(y_{p,r})$ on every proposal. For a fixed fractional-loss table the rank-coin pairs are iid. Let μ be the representative's actual tail risk, and put $p = \mu/q$. Then $\Pr(B = 1) = p$. Define the cumulative loss curve and its empirical version by

$$F_0(s) = \mathbb{E}[B\mathbf{1}\{\mathrm{rank} \le s\}], \qquad \widehat{F}_t(s) = \frac{1}{t} \sum_{i=1}^{t} B_i \mathbf{1}\{\mathrm{rank}(I_i) \le s\}. \tag{C39}$$

Residual concentration. If $\mu \le 2\tau$, then for every $x > 0$, except with probability at most $4e^{-x}$,

$$q \sup_s |\widehat{F}_t(s) - F_0(s)| \le r(t, x), \qquad r(t, x) = 4\sqrt{\frac{q\tau x}{t}} + \frac{2qx}{t}. \tag{C40}$$

To see this, put $M=\sum_i B_i$ and $v=x/t$. Binomial Bernstein [12] gives, except with probability $2e^{-x}$,

$$|M/t-p|\le\sqrt{2pv}+\tfrac{2v}{3},\qquad M/t\le 2p+\tfrac{7v}{6}.$$

Conditional on $M=s>0$, the successful ranks are iid from their conditional rank distribution [12]. Dvoretzky--Kiefer--Wolfowitz--Massart [13] bounds its empirical CDF error by $\sqrt{x/(2s)}$, except with conditional probability $2e^{-x}$. If $M=0$, that contribution is zero. Hence

$$\begin{aligned}\sup_s|\widehat{F}_t(s)-F_0(s)|&\le|M/t-p|+\sqrt{(M/t)v/2}\\&\le(\sqrt{2}+1)\sqrt{pv}+(\tfrac{2}{3}+\sqrt{\tfrac{7}{12}})v\\&\le 4\sqrt{\tfrac{\tau x}{qt}}+\tfrac{2x}{t},\end{aligned}$$

where $p\le 2\tau/q$. Multiplying by q proves (C40), including $p=0$.

For each target estimate its tail risk by

$$\widehat{R}^{\mathrm{tail}}_{n,t}=\tfrac{q}{t}\sum_{i=1}^{t}h_n(\mathrm{rank}(I_i))B_i. \tag{C41}$$

Summation by parts for the nonincreasing functions $h_n\in[0,4]$ gives, on the one event (C40),

$$\max_{n\in\mathcal{K}}|\widehat{R}^{\mathrm{tail}}_{n,t}-R^{\mathrm{tail}}_n|\le 4r(t,x). \tag{C42}$$

No union bound over targets is used: the nonnegative successive differences of each h_n sum to at most 4.

Use stages $j=1,2,\dots$ with

$$\begin{aligned}\alpha_j=\delta 2^{-j},\ x_j=\log(4/\alpha_j)=L+(j+2)\log 2,\\ A=\tfrac{4096q\tau}{\varepsilon^2},\ t_j=\lceil Ax_j\rceil,\ r_j=r(t_j,x_j).\end{aligned} \tag{C43}$$

Pass at the first stage at which

$$b_n+\widehat{R}^{\mathrm{tail}}_{n,t_j}+4r_j\le\tau\quad\text{for every } n\in\mathcal{K}. \tag{C44}$$

For global validity, consider any unsafe fixed table. If $\mu\le 2\tau$, passage requires failure of (C42), so has probability at most α_j at stage j. If $\mu>2\tau$, the declared representative a is itself unsafe, and passage implies $qM/t_j\le\tau<\mu/2$. A Bernoulli lower-tail Chernoff bound gives

$$\Pr(\text{pass at stage } j)\le e^{-t_j\mu/(8q)}\le e^{-t_j\tau/(4q)}\le e^{-x_j}\le\alpha_j. \tag{C45}$$

The last step uses $t_j\ge 4qx_j/\tau$, which follows from the standing parameter conditions and (C43). Thus the probability of ever passing an unsafe table is at most $\sum_j\alpha_j=\delta$. Neither case assumes the supplied margin is true of that table.

On a table in $\mathcal{Y}_\varepsilon$, $\mu\le\tau$ and

$$r_j\le\tfrac{\varepsilon}{16}+\tfrac{\varepsilon^2}{2048\tau}<\tfrac{\varepsilon}{8}. \tag{C46}$$

On (C42), each expression in (C44) is at most $\tau-\varepsilon+8r_j<\tau$. Therefore the probability of not passing by stage j is at most α_j. If U counts the uncapped proposal draws until passage, then

$$\begin{aligned}\mathbb{E}U &\le t_1+\sum_{j\ge 1}(t_{j+1}-t_j)\alpha_j\\ &\le A(L+3.25\log 2)+1.25\\ &\le 3AL=12288Hq.\end{aligned} \tag{C47}$$

For the last inequality, $L\ge 2\log 2$ leaves at least $0.75A\log 2$ to absorb 1.25; and $q>\varepsilon$ gives $A>4096\tau/\varepsilon\ge 65536$.

Cap tail proposals at the number D_k of remaining cells in the union of positive target supports. If passage has not occurred, query every remaining unknown support cell and evaluate all risks exactly. This adds no false certificate and ensures finite certification for every safe table. Coupled with the uncapped sequence, the distinct-label cost beyond the prefix is at most $\min\{U,D_k\}$: if $U<D_k$ it is at most U, and otherwise the census costs at most D_k. Thus the margin-class expected cost is at most

$$Pk+12288Hq_k\le 12288\{Pk+HZ_{\mathrm{tail}}(k)\}. \tag{C48}$$

The same displayed bound holds when $q_k\le\varepsilon$ by the deterministic prefix rule. Optimize over k using (C4), and also choose the deterministic k_ε auditor when its bound is smaller. This gives

$$C^*(\varepsilon)\le\min\{Pk_\varepsilon,12288G\}\le 12288F, \tag{C49}$$

completing the theorem.

References

[1] Larry Mai, with substantial Codex assistance. *ai-goodhart*, public v0.14 research artifact (2026), frozen source at `765ac51`. Source and evidence. The principal statements are in all-grid minimax and cached likelihood audits. The secondary finite example uses the separate placement and checkpoint proofs. The repository records exact-check protocols, saved outputs, source pins, rational callback contracts, and contribution disclosures.

[2] Reiichiro Nakano et al. *WebGPT: Browser-assisted question-answering with human feedback*. arXiv:2112.09332, first posted 2021; inspected version v3 (2022), Appendix I, "Predicting rejection sampling performance." Primary preprint. Its all-subsets evaluation uses binomial rank weights for finite-pool rejection sampling.

[3] Ricardo Fitas. *The geometry of AI validation: From structural blindness to reusable audits*. arXiv:2608.21496v2 (2026), Section 2.4 and Supplements S2, S5, and S7.2. Primary preprint. Relevant antecedents include envelope acquisition, the continuous-rank normalizer, and best-of-N acquisition bounds. The finite-table all-pass objective here must be compared at the level of its quantifiers.

[4] Sohail (Neel) Sarkar and Shakuntala Baichoo. *Cheap to Draw, Expensive to Trust: Certifying Test-Time Scaling Curves*. arXiv:2609.40190v1 (2026), Section 7.2 and Appendix C.3. Primary preprint. These passages give deterministic-grade reuse and geometric-band rare-risk information arguments. The paper also addresses simultaneous curve certification; that is a different output from one all-pass threshold assertion.

[5] Nick Duffield, Carsten Lund, and Mikkel Thorup. *Priority sampling estimating arbitrary subset sums*. arXiv:cs/0509026v1 (2005), Sections 3-3.1. Primary preprint. The fixed-threshold scheme and its variance optimization use clipped inclusion probabilities. Their expected sample-size functional supplies the algebraic antecedent of the clipped sum in this manuscript; the optimization objective there is distinct from adaptive certification cost.

[6] Tony Wirth. *The Query Complexity of Estimating Weighted Averages*. Google TechTalk delivered 4 February 2010, posted 16 February 2010, crediting joint work with Amit Chakrabarti, Venkatesan Guruswami, and Andrew Wirth. Official primary talk. The inspected slide at 14:32 shows heavy-set inspection and normalized residual sampling, with a squared residual-mass cost and explicit confidence dependence. The slide at 23:12 uses deterministic prefix queries until the remaining geometric-weight mass is below half the desired additive accuracy. This talk is an antecedent for both mechanisms, not a substitute for checking every quantifier of the later journal theorem [11].

[7] Lasse Fischer and Aaditya Ramdas. *Improving Wald's (approximate) sequential probability ratio test by avoiding overshoot*. arXiv:2410.16076v4 (2025), Sections 1-2. Primary preprint. The paper describes the classical conservative power-one likelihood-ratio test, its Ville-inequality justification, and the distinction between valid conservative thresholds and exact sample-size optimality.

[8] Ran Canetti, Guy Even, and Oded Goldreich. *Lower Bounds for Sampling Algorithms for Estimating the Average*. Information Processing Letters 53(1), 17-25 (1995); inspected author manuscript dated 6 October 1994, Sections 2-3. Primary author manuscript. Adaptive fixed-function queries, hard average classes, and expected-query arguments precede this work. The inspected finite-size regime and unweighted estimation objective differ from the present theorem.

[9] Shubhanshu Shekhar, Ziyu Xu, Zachary Lipton, Pierre Liang, and Aaditya Ramdas. *Risk-limiting financial audits via weighted sampling without replacement*. UAI, PMLR 216:1932-1941 (2023), Section 2, Definition 2, and Theorem 1. Primary proceedings page. Weighted finite-population confidence sequences, predictable acquisition, and control variates for side information are established antecedents.

[10] Jacob V. Spertus, Mayuri Sridhar, and Philip Stark. *Sequential stratified inference for the mean*. arXiv:2409.06680v5 (2026), Sections 2.1-2.2 and 3.2, Lemma 1, and Appendix C. Primary preprint. The inspected treatment includes finite populations, adaptive stratum selection, composite-mean testing, and the difference between logical stopping time and actual sample count.

[11] Amit Chakrabarti, Venkatesan Guruswami, Andrew Wirth, and Anthony Wirth. *The query complexity of estimating weighted averages*. Acta Informatica 48:417-426 (2011). DOI: 10.1007/s00236-011-0145-8. Primary publisher record. The publisher abstract and references were accessible; the final paper's full proof was not obtained. The earlier primary talk [6] establishes specific algorithmic antecedents, while the full finite-size, confidence-dependent, safe-side expected-cost comparison remains open.

[12] Stéphane Lhaut, Anne Sabourin, and Johan Segers. *Uniform concentration bounds for frequencies of rare events*. arXiv:2110.05826v2 (2022), Lemma 2.1 and equation (2.1). Primary preprint. Conditioning on the rare-region count and then combining conditional concentration with binomial Bernstein is an established device.

[13] Henry W. J. Reeve. *A short proof of the Dvoretzky-Kiefer-Wolfowitz-Massart inequality*. arXiv:2403.16651v1 (2024), Theorem 1 and Corollaries 1-2. Primary preprint. This provides a modern proof and attribution for empirical-CDF concentration; the manuscript uses the classical two-sided bound.

[14] Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. *Exponential line-crossing inequalities*. arXiv:1808.03204v4 (2019), Section 1.2. Published as *Time-uniform Chernoff bounds via nonnegative supermartingales* (2020). Primary preprint. The stated Ville inequality for a nonnegative supermartingale is the classical maximal inequality applied in Appendix A.3.